Design and Implementation of Low Power 8-bit SAR ADC

Major Project Report

Submitted in Partial Fulfillment of the Requirements for the Degree of

MASTER OF TECHNOLOGY IN ELECTRONICS AND COMMUNICATION ENGINEERING

by

Moksh Prajapati

(24MRV011)

Department of Electronics and Communication Engineering
Institute of Technology
Nirma University, Ahmedabad 382 481

May 2026

Design and Implementation of Low Power 8-bit SAR ADC

Major Project Report

Submitted in Partial Fulfillment of the Requirements for the Degree of

MASTER OF TECHNOLOGY IN ELECTRONICS AND COMMUNICATION ENGINEERING

by

Moksh Prajapati (24MRV011)

Under the Guidance of

Dr. Piyush Bhatasana

Department of Electronics and Communication Engineering
Institute of Technology
Nirma University, Ahmedabad 382 481

May 2026

Certificate

This is to certify that the Major Project Report entitled "Design and Implementation of Low Power 8-bit SAR ADC" submitted by Mr. Moksh Prajapati (24MRV011) towards the partial fulfillment of the requirements for the award of degree in Masters of Technology in the field of Electronics & Communication Engineering of Nirma University is the record of work carried out by him under our supervision and guidance. The work submitted has in our opinion reached a level required for being accepted for examination. The results embodied in this major project work to the best of our knowledge have not been submitted to any other University or Institution for award of any degree or diploma.

Date:

Project - Guide
Dr. Piyush Bhatasana
Assistant Professor, EC Department,
Institute of Technology,
Nirma University,
Ahmedabad 382481

Program Coordinator,
Dr. Usha Mehta
Professor, EC Department,
Institute of Technology,
Nirma University,
Ahmedabad 382481

Head of Department
Dr. Nagendra Gajjar
Department of Electronics &
Communication Engineering,
Institute of Technology,
Nirma University,
Ahmedabad 382481

Director
Dr. Himanshu Soni
School of Technology,
Institute of Technology,
Nirma University,
Ahmedabad 382481

Undertaking for Originality of the Work

I, Moksh Prajapati (24MRV011) give undertaking that the Major Project entitled "Design and Implementation of Low Power 8-bit SAR ADC" submitted by me towards the partial fulfillment of the requirements for the degree of Master of Technology in Electronics and Communication Engineering, VLSI Design of Nirma University, Ahmedabad 382 481, is the original work carried out by me and I give assurance that no attempt of plagiarism has been made. I understand that in the event of any similarity found subsequently with any other published work or any project report elsewhere; it will result in severe disciplinary action.

Signature of the student: ____________________

Date: _______________

Place: _______________

Endorsed By:

(Signature of Internal Guide)

ACKNOWLEDGEMENT

The journey of completing this minor project on Sucessive Approximation Register ADC has been enriched by the support and guidance of many individuals, whose contributions we deeply appreciate.

We would like to express our sincere gratitude to our external mentor, Dr. Piyush Bhatasna for their consistent support, expert advice, and valuable feedback throughout the course of this work. Their mentorship played a vital role in enhancing our understanding of the subject and guiding us through the technical challenges of the project. Their encouragement has been a constant source of motivation.

We are also thankful to our Head of Department, Dr. Nagendra Gajjar, for providing a stimulating academic environment and for her continued encouragement, which greatly contributed to the successful execution of this project.

Our appreciation also extends to all the faculty members of the department for their dedication to teaching and their willingness to help whenever needed. Their support has been instrumental in our academic development and in the successful completion of this work.

This project reflects the collaborative effort of everyone who contributed directly or indirectly, and we are truly grateful for their involvement.

ABSTRACT

This project focuses on the design and simulation of an 8-bit Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC) using the 45 nm CMOS technology, targeting low-power and medium-resolution applications such as biomedical instrumentation, portable electronics, and IoT sensor platforms. SAR ADCs have become increasingly attractive due to their excellent balance between power efficiency, silicon area, and moderate sampling speed. In this work, all major functional blocks—including the Sample-and-Hold circuit, dynamic comparator, SAR logic, capacitive DAC, and output register were designed, analyzed, and individually verified before integrating into a complete ADC architecture. The objective was to achieve reliable bit-by-bit approximation while minimizing energy consumption and ensuring stable timing operation across all blocks.

The Sample-and-Hold (S/H) circuit provides clean and linear sampling with minimal droop, ensuring accurate representation of the input before conversion. A low-power dynamic latch comparator is implemented to provide fast regeneration and robust decision-making with negligible static power consumption. The SAR control logic, designed using TSPC flip-flops, ensures correct bit sequencing, stable charge storage, and synchronized interactions with the DAC and comparator. The binary-weighted capacitive DAC exhibits monotonic behavior and demonstrates accurate charge redistribution consistent with theoretical SAR operation. Simulation waveforms validate correct switching behavior, low settling errors, and smooth interaction among all sub-blocks.

Overall, the integrated architecture achieves low power consumption, consistent 8-bit performance, and stable operation across the conversion cycle. The modular design methodology reinforces the suitability of SAR ADCs for energy-constrained environments. This work establishes a strong foundation for future enhancements, including differential DAC structures, improved timing control, and extension to higher resolutions for advanced mixed-signal applications.

TABLE OF CONTENTS

List of Figures

List of Tables

NOMENCLATURE

ADC .... Analog-to-Digital

DAC .... Digital-to-Analog

ENOB .... Effective number of bits

SINAD .... Signal-to-Noise And Distortion ratio

INL .... Integral Non-Linearity

DNL .... Differential Non-Linearity

Chapter 1

INTRODUCTION

1.1 Background

Every digital system that touches the physical world needs something to convert analog signals into numbers. The SAR ADC has become the default for that job in most modern designs — not because it's the fastest or highest resolution option, but because it sits in a useful middle ground that few architectures match. That middle ground is increasingly where the industry lives. Biomedical implants, IoT sensors, and portable devices all need converters that run on small power budgets without giving up too much accuracy. A SAR ADC fits by doing something computationally cheap: binary search. Each clock cycle it makes a guess, checks high or low against the actual input, and narrows down. A handful of cycles later it has a result. The hardware is minimal by design. A sample-and-hold circuit freezes the input voltage, a capacitive DAC generates reference levels for each comparison, a single comparator checks each guess, and a small logic block keeps track of where the search is. No active amplifiers the circuit mostly redistributes charge passively between capacitors. That's why it scales well with newer semiconductor processes instead of fighting them.

1.2 Motivation

The relentless push toward compact, energy-efficient electronics has exposed the limitations of traditional ADC architectures. While Flash and Pipeline ADCs offer blistering throughput, their reliance on massive comparator arrays and active residue amplifiers incurs severe power and area penalties, rendering them entirely unsuited for stringent energy-harvesting or battery-operated environments. Conversely, architectures that prioritize extreme accuracy, such as Sigma-Delta ($\Sigma\Delta$) converters, are fundamentally handicapped by limited bandwidth and the heavy power overhead of complex digital decimation filters. This architectural gap creates a compelling motivation to aggressively optimize the SAR ADC. While the baseline SAR architecture is inherently efficient, standard implementations still suffer from dynamic power waste

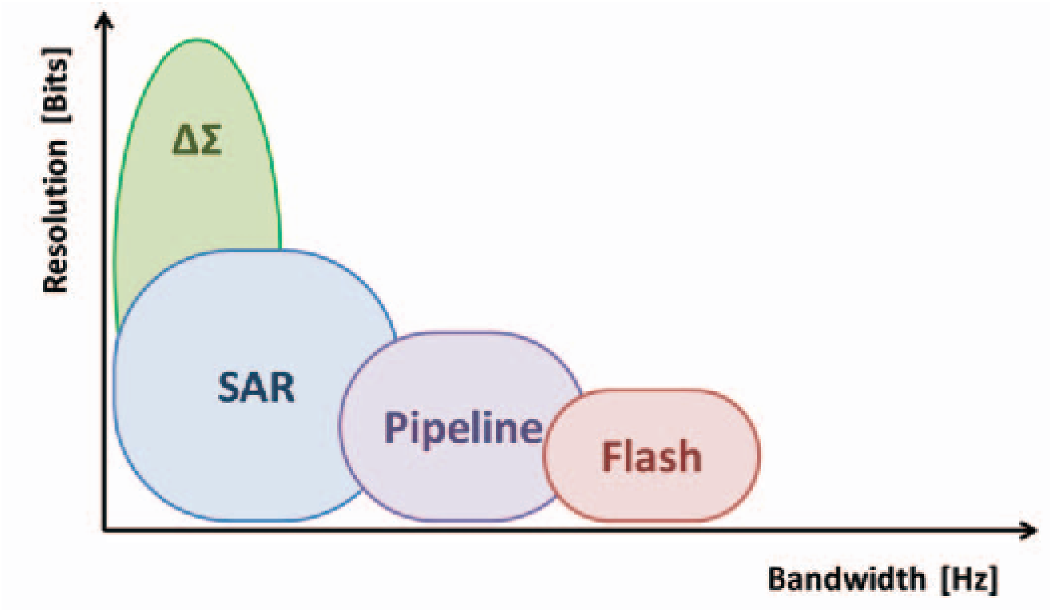

Figure 1: Popular ADC Architecture's Resolutions VS Bandwidth

during capacitive switching and tracking non-linearities at higher frequencies. Pushing the boundaries of the Walden Figure of Merit (FoM) in a deep sub-micron process such as 45nm CMOS requires rethinking the internal building blocks. By addressing the parasitic limitations of the DAC and the speed-versus-noise trade-offs in the dynamic comparator, there is a clear pathway to designing a converter that achieves the strict linearity and resolution required by next-generation smart sensors without violating their tight thermal and power constraints.

1.3 Problem Statement

While SAR ADCs are well-suited for lower- and moderate-resolution applications, several key challenges limit their performance in advanced designs. One major issue is the capacitive DAC architecture, where binary-weighted arrays lead to exponential growth in capacitance with increasing resolution. This not only increases chip area and power consumption but also introduces capacitor mismatch, degrading linearity and accuracy. The comparator, a critical analog component, also contributes significantly to power consumption and is vulnerable to kickback noise and offset error. Though techniques like sub-threshold operation reduce power, they often compromise speed and accuracy, especially at higher sampling rates.

In addition, the SAR logic block becomes increasingly complex as resolution increases. Managing timing (setup/hold) and synchronization across flip-flops and control signals is essential but difficult, particularly in asynchronous designs or when using low supply voltages.

Finally, process scaling and low-voltage operation (e.g., sub-1V) further strain analog performance, as key components like the comparator and DAC become more sensitive to variations. These challenges collectively hinder efforts to design SAR ADCs that are simultaneously high speed, high resolution, low power, and compact, necessitating optimized architecture and circuit-level innovations.

1.4 Objectives

The primary objective of this work is to design and implement a low-power, high-efficiency SAR ADC architecture that meets the performance demands of modern electronics systems, particularly in portable and biomedical applications. The design aims to balance power, speed, resolution, and area efficiency through architecture and circuit-level optimizations.

The specific objectives of the project are

- To design a SAR ADC architecture using CMOS 45 nm technology that operates efficiently at low supply voltage (1 V - 1.2 V), making it suitable for power-sensitive applications.
- To implement a power-optimized comparator, such as a double-tail latch or sub-threshold design, that minimizes energy consumption while maintaining high-speed operation and low offset.
- To develop an efficient DAC, utilizing split capacitor or R-2R ladder architecture, to reduce total capacitance, improve matching, and lower switching power.
- To construct SAR logic using D flip-flop-based shift registers that offer simple control and reliable bit-by-bit decision-making during analog-to-digital conversion.
- To achieve a compact and scalable design that is easily adaptable for resolutions ranging from 6 to 10 bits, enabling its use in a wide variety of embedded systems and sensor interfaces.

1.5 Approach

We're taking a step-by-step, bottom-up approach to designing this SAR ADC using Cadence Virtuoso's 45nm library. Instead of building it all at once, we design and test each main piece individually to ensure everything works perfectly before integration. To hit our low-power and moderate-resolution goals, we chose highly efficient components: a double-tail latch comparator to balance speed and energy, a split-capacitor (or R-2R) DAC to reduce switching power, and D flip-flop SAR logic to keep the bit-by-bit approximation perfectly synced. Once these individual blocks are working, we wire them together and simulate the entire system to verify our final power, speed, and linearity (INL/DNL) metrics.

1.6 Scope of the Project

This research focuses on the comprehensive circuit-level design and simulation of a low-power Successive Approximation Register (SAR) ADC utilizing the GPDK 45nm CMOS process technology. The design specifically targets an 8-bit resolution architecture, optimized for the moderate sampling speeds required by biomedical monitoring systems and ambient IoT sensor nodes.

The scope encompasses the detailed schematic capture, transistor sizing, and transient analysis of all fundamental sub-blocks. This includes the design of a highly linear sampling network, an energy-efficient capacitive DAC, a low-offset dynamic comparator, and the supporting digital state machine. The project will critically evaluate key static and dynamic performance metrics specifically Differential Non-Linearity (DNL), Integral Non-Linearity (INL), Signal-to-Noise Ratio (SNR), and the Effective Number of Bits (ENOB). Through iterative simulation using industry-standard EDA tools, the complete integrated architecture will be heavily optimized to minimize overall power consumption while maintaining stringent signal integrity across the designated analog input range.

Chapter 2

LITERATURE REVIEW

The design of Successive Approximation Register Analog-to-Digital Converters has been an active area of research dur to their wide applicability in low-power, moderate-speed systmes such as biomedical devices, wireless sensor nodes, and mobile electronics. The core problem in SAR ADC design revolves around achieving high resolution, fast sampling speed, and low power consmpution within a minimal silicon area. Various architectural and circuit-level innovation have been proposed over time to addrress these trade-offs. This section provides a comprehensive review of the significant approaches developed to solve these challenges.

2.1 A Low-Noise 10-bit SAR ADC Using Chopper Comparator for Biomedical [1]

This paper addresses one of the most stubborn physical limitations in modern biomedical sensor design: the inescapable reality of low-frequency noise. When designing sensors to monitor biological signals like electrocardiograms (ECGs) or electroencephalograms (EEGs), engineers are forced to work in the sub-kilohertz frequency range. Unfortunately, standard sub-micron CMOS transistors are plagued by flicker noise (also known as $1/f$ noise), which grows exponentially stronger at these low frequencies, effectively drowning out the delicate biological signals. Furthermore, slight manufacturing variations cause threshold mismatches, creating a static DC offset that corrupts the digitized data. Instead of utilizing the brute-force method of simply making the transistors massive to reduce noise which would violate the strict area and power constraints of wearable implants this paper ingeniously employs a chopper stabilization technique right inside the comparator. By strategically placing cross-coupled switches before and after the amplification stage, the comparator modulates the incoming biological signal up to a much higher chopping frequency. The analog circuitry then processes the signal alongside its own low-frequency flicker noise. When a second set of switches demodulates the signal back down to baseband, it simultaneously forces the comparator's own noise and offset up into the high-frequency spectrum, where it is effortlessly stripped away by digital decimation filters. This elegant architectural maneuver allows the ADC to achieve a pristine 59.41 dB Signal-to-Noise Ratio (SNR) in a 0.18 µm process, proving that high clinical accuracy can be achieved without sacrificing the stringent energy requirements of battery-powered biomedical implants.

2.2 A 10-bit 160 MS/s Asynchronous SAR ADC Design [2]

Pushing a SAR ADC to operate at 160 Mega-Samples per second (MS/s) completely shatters the traditional boundaries of this architecture, and this paper demonstrates exactly how to navigate the severe timing bottlenecks that arise at such extreme speeds. In a standard synchronous design, achieving 160 MS/s would require an external clock running in the gigahertz range to step through the successive approximation algorithm. Distributing a clock that fast across a silicon layout consumes an enormous amount of dynamic power and introduces fatal clock skew. Jiaqi Wang's design brilliantly bypasses this by discarding the global clock entirely in favor of an asynchronous, self-timed logic loop. The system relies on a customized dynamic comparator that generates its own "Ready" pulse the microsecond it finishes resolving a bit, instantly triggering the digital logic to shift to the next phase. This entirely eliminates the wasted "wait time" normally allocated for worst-case settling scenarios. Furthermore, at these speeds, charging and discharging massive capacitor arrays drains the reference buffer rapidly. To combat this, the author implements a split-capacitor array coupled with a monotonic switching scheme. By only switching capacitors downward to ground rather than charging them up to the reference voltage, the design slashes the switching energy by over 80%. This paper is a masterclass in co-optimizing both the digital backend and the analog frontend to achieve aggressive throughput without melting the silicon.

2.3 An Ultra-Low-Power SAR ADC with an Area-Efficient DAC Architecture [3]

While many designers chase blistering speed, this paper pivots entirely to the pursuit of extreme miniaturization and nanowatt power consumption, targeting the burgeoning field of radio-frequency identification (RFID) and energy-harvesting smart dust. The fundamental enemy of area efficiency in a standard SAR ADC is the binary-weighted capacitive digital-to-analog converter (DAC). Because the capacitors must scale by powers of two, an 8-bit or 10-bit array grows exponentially large, dominating the physical layout and creating massive parasitic leakage paths. Kamalinejad and his team propose a radical, highly unorthodox architectural departure: they entirely abandon the traditional binary array. Instead, their proposed DAC utilizes only four minimum-sized unit capacitors. To accomplish the successive division of the reference voltage without a massive capacitor bank, they incorporate two rail-to-rail, low-power unity-gain buffers to actively drive the nodes and hold the charge. While integrating active operational amplifiers into a traditionally passive SAR network seems counterintuitive for power saving, the massive reduction in switched capacitance ends up winning the energy trade-off. Because the capacitance is so incredibly small, the dynamic switching power drops through the floor, allowing the entire chip to consume a microscopic 290 nanowatts. The strict trade-off, however, is speed; the settling time of the active buffers limits the conversion rate to a mere 25 kS/s, firmly positioning this design as a specialized solution for ultra-slow, ultra-low-power environmental monitoring.

2.4 Design of an Internal Asynchronous 11-bit SAR ADC for Biomedical Wearable Applications [4]

This paper tackles the unique environmental hazards of embedding high-precision ADCs directly into wearable consumer electronics. Unlike laboratory environments, wearables are subjected to massive amounts of external electromagnetic interference, fluctuating battery supply voltages, and severe temperature variations. Shiue and his team address these hazards by implementing a fully differential analog front-end. By duplicating the sampling pathways, any common-mode noise whether it is interference coupling through the skin or digital switching noise leaking through the silicon substrate is mathematically cancelled out when the differential signals are subtracted at the comparator's input. Additionally, to maximize battery life, the design incorporates an internal asynchronous clock generator that allows the ADC to aggressively scale its power consumption based on the required sampling rate. If the wearable device drops from active monitoring into a low-power standby mode, the asynchronous loop naturally stretches out, drastically dropping the dynamic power consumption without needing an external microcontroller to reprogram the clock tree. Operating at a low 1.2V supply and fabricated in a mature, cost-effective TSMC 0.18 µm CMOS technology, this 11-bit design represents a highly pragmatic, robust solution tailored specifically for the commercial realities of the wearable medical device market.

2.5 Design of Binary Architecture for Successive Approximation Analog-to-Digital Converter [5]

This research by Tejender Singh pushes the SAR architecture into physical territory traditionally reserved exclusively for power-hungry Pipeline ADCs. Achieving 16 bits of resolution at a staggering 500 MS/s in a deep sub-micron 45nm process is an exercise in managing extreme physical non-idealities. At 16 bits, the Least Significant Bit (LSB) is vanishingly small, meaning thermal noise (kT/C) and harmonic distortion must be kept to near-zero levels. The focal point of this design is its rigorous approach to the sample-and-hold (S/H) circuitry. Standard CMOS transmission gates suffer from signal-dependent on-resistance, which distorts high-frequency input signals before the conversion even begins. To survive at 500 MS/s, this design heavily leverages bootstrapped switching architectures. By using a pre-charged capacitor to act as a floating battery, the circuit rigidly locks the gate-to-source voltage (V_{GS}) of the sampling transistor to the exact supply voltage, completely independent of the analog input swing. This guarantees a perfectly flat, distortion-free resistance profile across the entire tracking phase. Furthermore, operating in 45nm requires aggressive mitigation of leakage currents and severe attention to routing parasitics, as even a few femtofarads of stray metal capacitance can destroy the matching of the 16-bit array. This paper highlights the absolute bleeding edge of what passive charge-redistribution architectures can achieve before requiring active residue amplification.

The comparator uses a three-stage structure consisting of a pre-amplifier, a regenerative latch, and an SR-latch output stage. The pre-amplifier provides initial gain and reduces offset, while the latch resolves small differential inputs quickly, and the SR latch stabilizes the digital output. The SAR logic employs SR-based D flip-flops to reduce switching power, making the overall ADC efficient for low-resolution, low-voltage applications.

2.6 Comparative Summary of Approaches

Work	**Technology**	**Resolution**	**Target Application**	**Unique feature**
Pham et al. (2024)	180 nm CMOS	10-bit	Biomedical / Clinical Sensors	Chopper comparator with RC-integrator to cancel offset ripple, compact 0.457 mm^2 layout.
Wang (2023)	40 nm CMOS	10-bit	High-Speed Data Links	160 MS/s operation using split–monotonic CDAC, bootstrapped sampling and asynchronous SAR logic.
Kamalinejad et al. (2011)	130 nm CMOS	8-bit	RFID / Smart Dust	Area-efficient 4-capacitor DAC with two unity-gain buffers; buffers reused for comparator preamp.
Shiue et al. (2024)	180 nm CMOS	11-bit	Wearable Medical Devices	Fully-differential SAR with internal asynchronous clock generator and monotonic switching for wearables.
Nitnaware et al. (2015)	90 nm CMOS	4-bit	High-Frequency Telecom	Binary CDAC that inherently performs sample-and-hold, plus 3-stage comparator (preamp + latch + SR) and SR D-FFs.

Table 1: Comparative summary of SAR ADC design approaches .

Chapter 3

Science Behind ADCs—From Flash to SAR

This chapter explores the fundamental principles, architecture, and performance metrics of Analog-to-Digital Converters. It presents a comparative overview of various ADC including flash, pipeline, sigma-delta with a focused analysis on SAR ADC design.

3.1 How does ADC works?

The real world doesn't care about round numbers. Temperature, sound, the voltage from a microphone these things change continuously. Draw them out and you get a smooth rolling curve. Computers can't work with that; they need discrete numbers, so something has to translate between the two. That's an ADC.

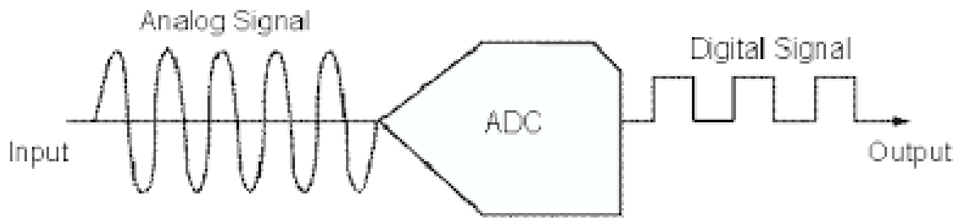

Figure 2: Basic block diagram of SAR ADC

First, it samples. Think of it like photographing a moving car you can't freeze every moment, so you just freeze it repeatedly. The ADC grabs the voltage at regular intervals and holds each value steady until the next snapshot arrives. The smooth curve turns into a stair-step pattern. But here's the thing: those frozen values can still be messy decimals. 3.14159 volts, for example. That's not useful. So the next step, quantization, rounds everything to the nearest predefined level same idea as a ruler with only inch marks. 4.2 becomes 4. 4.8 becomes 5. Every step snaps to something clean. Then encoding just writes those clean numbers out in binary. The math-heavy part is already done by then. Photograph the wave at regular intervals. Round the measurements. Convert to binary. That's the whole thing.

3.2 Flash ADC

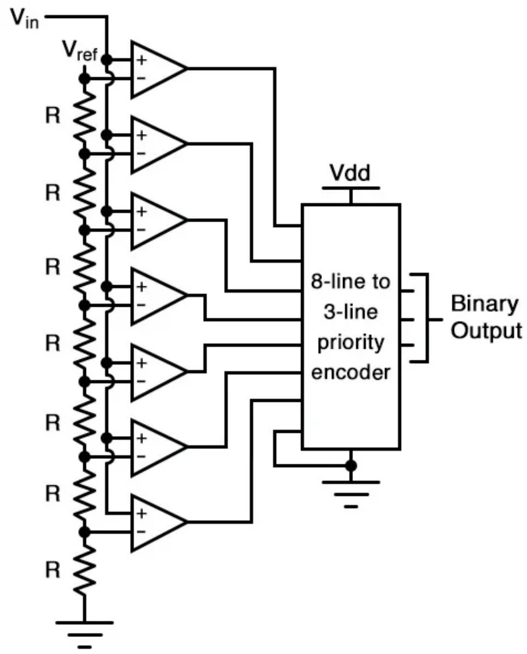

Figure 3: Basic Block diagram of Flash ADC

Flash ADCs don't do anything clever. They just throw hardware at the problem. Imagine every inch from floor to ceiling has its own motion sensor. Measuring your height doesn't require any calculation every sensor below your head trips, everything above stays quiet. Read where the line switches and you have your answer. A Flash ADC is that wall of sensors. A resistor ladder divides the reference voltage into every possible level, each with its own dedicated comparator. Feed in the analog input and all of them see it at once — comparators below the input voltage flip to 1, the rest stay 0, and a digital encoder converts the pattern to binary. One clock cycle, start to finish. No feedback, no timing phases, nothing iterative. This makes them extraordinarily fast. Oscilloscopes and gigahertz optical links use them because there's essentially no latency the answer is just there. The problem is that resolution gets expensive fast. Each additional bit doubles the comparator count: 8-bit needs 255, 10-bit needs 1,023. At some point the chip becomes physically impractical too large, too power-hungry, and the hundreds of comparators drawing current simultaneously start loading down the input signal they're supposed to be measuring cleanly.

3.3 Sigma-Delta ADC

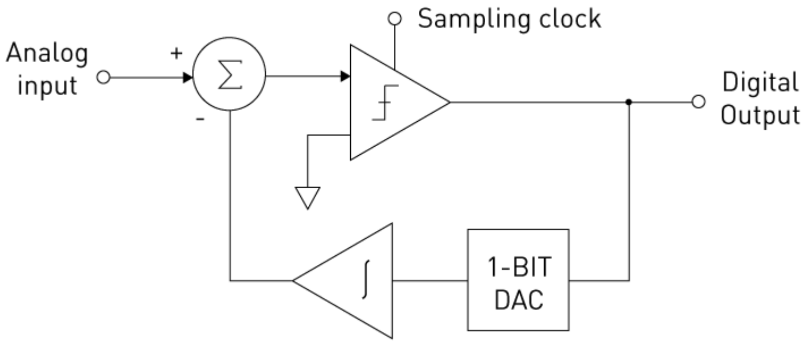

Figure 4: Basic Block diagram of Sigma Delta ADC

The teaspoon analogy still holds: sample fast enough and volume compensates for individual crudeness. But a Sigma-Delta ADC doesn't just oversample blindly — it runs a feedback loop at the same time. The circuit takes the analog input, subtracts its own last guess, and feeds the difference into an integrator. The integrator accumulates those errors and drives a 1-bit comparator that fires out a stream of 1s and 0s. High input voltage produces mostly 1s, low produces mostly 0s. The density encodes the signal. Each output gets fed back and subtracted from the next input, so errors get corrected rather than compounding. The part that makes this worth doing is noise shaping. The integrator doesn't just accumulate it pushes quantization noise up into high frequencies, away from the audio band where it would actually matter. A digital filter at the end cuts those frequencies off. What remains is clean enough to hit 20 to 24 bits of resolution, which is why Sigma-Delta is standard in audio equipment and precision measurement gear. It also doesn't demand tight analog tolerances. The resistors and capacitors don't need careful matching because the hard work happens in the digital filter, not the analog front end. The costs are speed and backend complexity. Hundreds or thousands of raw samples go in for every output word that comes out, so anything time-sensitive radar, video is off the table. And while the analog side is genuinely simple, the decimation filter that cleans up the noise and downsamples the stream is not small.

3.4 Dual Slope ADC

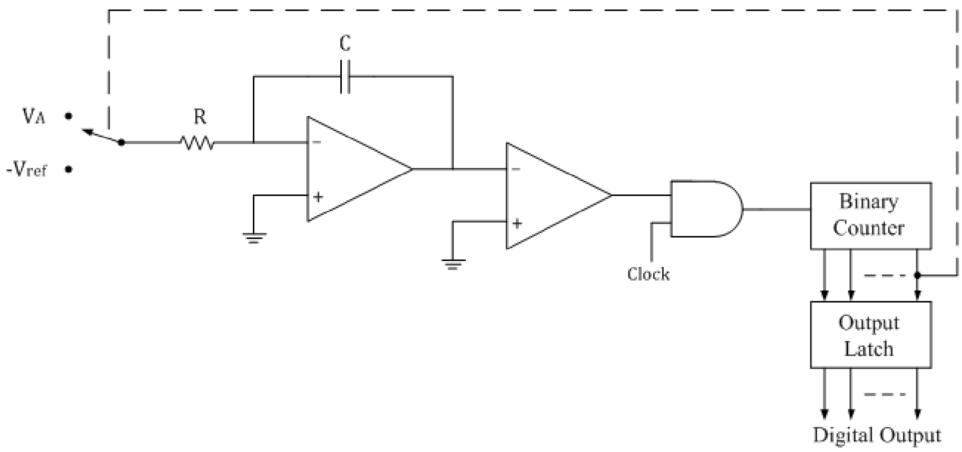

Figure 5: Basic Block diagram of Sigma Delta ADC

A Dual Slope ADC doesn't measure voltage directly it converts voltage into time, then measures the time. The analogy: fill a bucket with an unknown hose for exactly 10 seconds. Swap in a hose with a known flow rate, drain the bucket, and watch the clock. How long it takes to empty tells you how full it got, which tells you how hard the unknown hose was flowing. The circuit works the same way. First, the unknown input voltage drives an integrator for a fixed period a capacitor charges up. Then the input disconnects, a known reference voltage takes over at opposite polarity, and the capacitor drains back to zero while a counter ticks. Since you know exactly how long you charged and exactly what the reference is, the drain time works backward to give you the input voltage. Two things fall out of this design that are genuinely useful. Because the input gets averaged over the entire charge period rather than sampled at a point in time, high-frequency noise and 60Hz power line interference cancel themselves out. And because the same resistor and capacitor handle both phases, any imprecision in their values cancels out too you're not relying on their absolute accuracy, just their consistency. The problems are speed and resolution. Fully charging and then draining a capacitor for every reading is slow, so these ADCs end up in things like digital multimeters where measurement rate doesn't matter. Getting finer resolution means the counter has to tick faster during the drain and that hits a ceiling pretty quickly.

3.5 Successive Approximation Register ADC

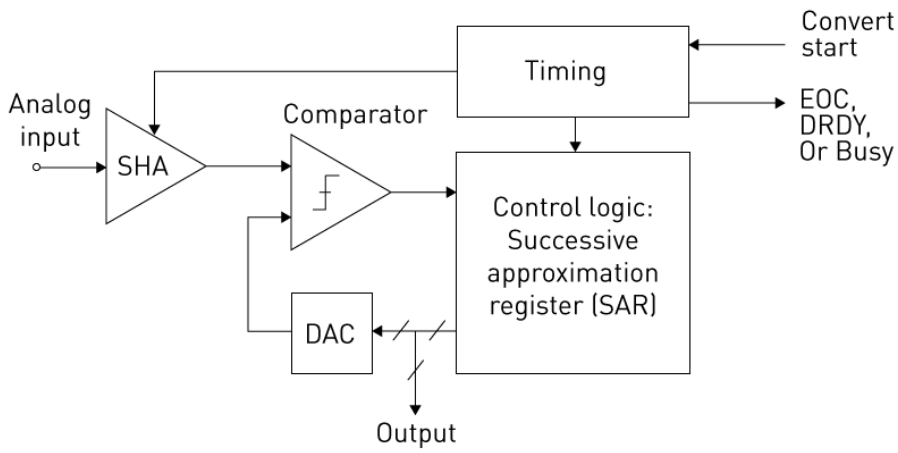

Figure 6: Basic block diagram of SAR ADC

A Successive Approximation Register (SAR) ADC is a data-conversion architecture designed to obtain a digital representation of an analog input by resolving one bit at a time. Unlike converter types that operate fully in parallel or rely on oversampling, the SAR approach uses a stepwise comparison strategy that allows it to reach a precise result with minimal circuitry. This makes SAR converters especially appealing for systems where energy budget, silicon area, and moderate speed requirements dominate the design constraints.

At the heart of the SAR ADC is a feedback loop that tests a sequence of digital codes generated by the SAR logic. Each of these codes corresponds to a voltage produced by the DAC. During a conversion cycle, the ADC temporarily stores the input on a sampling element and then begins a binary search. The logic initially assumes the input might lie in the upper half of the full-scale range, so the most significant bit (MSB) is tentatively set to '1'. The DAC generates the corresponding reference level, and the comparator checks whether the stored input is greater or smaller than this trial value. Based on the comparator output, the MSB is either confirmed or cleared.

Once the first bit is resolved, the SAR logic moves to the next bit with the previously determined value serving as a boundary. Each comparison halves the remaining search interval, gradually tightening the estimate of the input voltage. This continues until all bits have been processed, resulting in a digital word that represents the closest achievable approximation for the sampled input. Because only one comparison is carried out per bit, the power consumption remains significantly lower than architectures requiring multiple comparators or large analog gain stages.

The strength of the SAR architecture lies in this iterative but deterministic procedure. It avoids the continuous-time operation of sigma-delta converters and does not require the wide comparator arrays used in flash ADCs. The overall structure comprising a single comparator, a capacitor-based DAC, sampling circuitry, and simple digital control allows the converter to scale efficiently with technology nodes and enables operation at low supply voltages. As a result, SAR ADCs are widely adopted for embedded sensing, portable devices, and biomedical measurement systems, where a balance among speed, energy efficiency, and resolution is essential.

3.6 Working Principle of SAR Register

The SAR ADC is a widely used architecture that employs a binary search algorithm to efficient convert an analog input voltage into its corresponding digital representation. This converstion method is based on a step-by-step approximation of the input signal, starting from the MSB and progressing sequentially towards the LSB. A typical SAR ADC consist of four main component: a sample-and-hold circuit, a comparator, a digital-to-analog converter, and the SAR control logic. During each conversion cycle, the SAR logic (TSPC DFF) sets one bit to '1' and, using the DAC, generates a corresponding trail voltage. This voltage is then comparedwith the held analog input by the comparator. Depending on the result of the comparison, the bit is either confirmed or reset to '0'. This iterative process continues until all bits in the digital output have been resolved. The final digital value stored in the SAR register closely approximates the original analog input. SAR ADCs are known for their excellent power efficiency, compact design, and moderate speed,which makes them highly useful in compact electronics, medical monitoring systems, and various embedded sensing platforms where low power and reliable performance are essential for the basic usage.

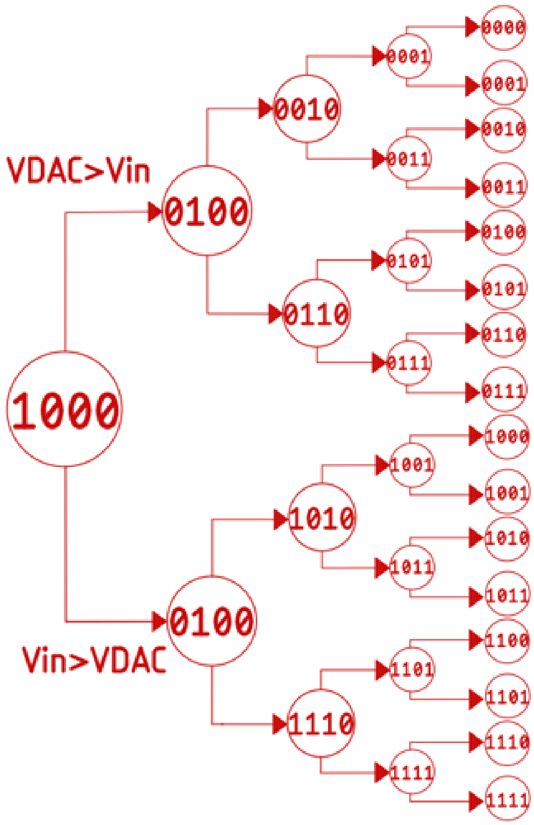

Figure 7: Binary Search Algorithm

Step-by-Step working principle

1. **Sampling Phase:**

   - The input analog voltage is sampled and held constant by a sample-and-hold circuit.

2. **Initialization:**

   - All bits in the SAR register are initially set to '0'.
   - The MSB is set to '1' to begin the binary search.

3. **First Comparison (MSB):**

   - The SAR logicnsets the MSB to '1' and sends the digital code to the DAC.
   - The DAC converts this code to an analog trail voltage.
   - The comparator compares the DAC output with the input voltage.
   - If $V_{in} > V_{DAC}$, the bit is retained; elese. it is reset to '0'.

4. **Successive Comparisons:**

    - The next bit is set to '1', and a new trail volatge is generated.
    - The comparator check again, and the SAR logic updates the bit accordingly.
    - This process continues for all bits, from MSB to LSB.

5. **Final Output:**

    - After all bits have been tested and fixed, the SAR register holds the final digital value.
    - This digital output is the closest binary approximation of the input analog voltage.

6. **Conversion Complete:**

    - The digital output can now be used by digital systems for further processing.

3.7 Parameters of ADC

3.7.1 Resolution

Resolution is the "marketing number." It is the sheer number of distinct digital levels the ADC is theoretically capable of producing. Think of it as the number of pixels on a TV screen more pixels means you can display finer, smoother details.

- The Math: An N-bit ADC slices the voltage range into 2^N discrete levels. For example, a 8-bit ADC has $2^8 = 256$ levels, while a 16-bit ADC has a massive 65,536 levels.

3.7.2 Sampling Rate (Sampling Frequency)

If resolution is how finely you can see, the sampling rate is how fast you can blink. It is the absolute speed limit of the ADC, telling you how many individual analog snapshots it can convert into digital words every single second. It is usually measured in Mega-Samples per second (MS/s) or Giga-Samples per second (GS/s).

- The Math Connection: The sampling rate is strictly bound by the Nyquist-Shannon sampling theorem. To successfully capture an analog signal without corrupting it (aliasing), the sampling rate must be strictly greater than twice the highest frequency of the input signal: $f_s \geq 2 \cdot f_{in,max}$.

3.7.3 Effective Number of Bits (ENOB)

ENOB tells how many bits of the ADC are actually useful in practice, accounting for real-world noise and imperfections. It is often slightly slower than the theoretical resolution.

3.7.4 Linearity (INL and DNL)

This is all about the physical accuracy of the "stair steps" inside the ADC. Imagine you buy a ruler. You expect every single millimeter mark to be exactly one millimeter apart. In an ADC, this ideal gap between digital codes is called the Least Significant Bit (LSB), calculated as $LSB = V_{ref}/2^N$.

- **INL (Integral Non-Linearity):** This is the cumulative effect of all those tiny DNL errors adding up. Going back to the ruler analogy: even if the individual millimeter marks look okay, the entire ruler might be physically bent. INL measures how far the actual ADC transfer curve deviates from a mathematically perfect, straight line.
- **DNL (Differential Non-Linearity):** This measures the error between adjacent steps. If one step is slightly wider or narrower than a perfect LSB, that's DNL. If the DNL ever hits -1 LSB, it means a step completely vanished, causing a "missing code."

Chapter 4

Hardware Design

4.1 Sample and Hold circuit

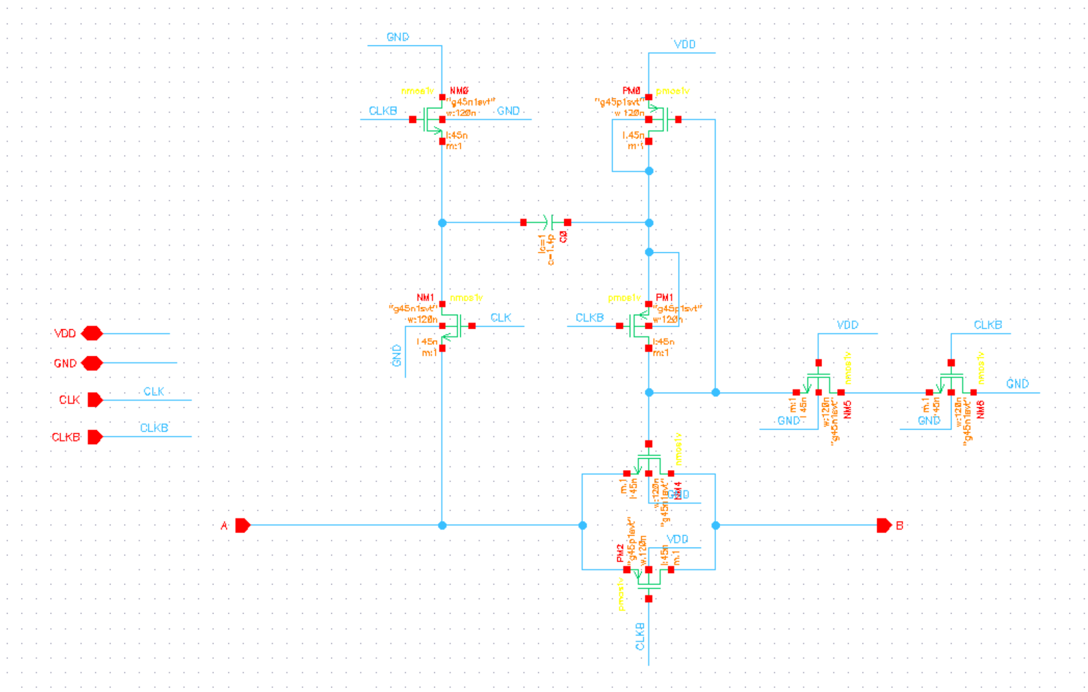

Figure 8: Sample & Hold Circuit

In my 8-bit SAR ADC, I decided to use a bootstrapped switch for the sample-and-hold circuit because, quite frankly, a standard NMOS switch just wouldn't give me the linearity I needed. If I had used a simple switch, its on-resistance (R_{on}) would have changed every time the input signal moved, creating a messy, signal-dependent distortion. By using the bootstrapped architecture shown in the schematic, I can keep the gate-to-source voltage (V_{GS}) of my main sampling switch perfectly constant, which keeps the resistance flat and the signal clean.Here is how I've set up the circuit to work across its two main phases:

- **Phase 1: The Hold (or Pre-charge) Phase:** When the clock (CK) is low, the circuit is in the "hold" phase. My main goal here is to keep the sampling switch (M_{sw}) completely off while getting the system ready for the next sample. During this time, the transistors M_3 and M_4 turn on, which connects my "battery" capacitor (C_{bat}) between the supply voltage (V_{DD}) and ground. Effectively, I'm "charging the battery" to V_{DD}. At the same time, the gate of my main switch is pulled to ground to make sure no signal leaks through to the sampling capacitor (C_s).

- **Phase 2: The Sampling (or Bootstrapping) Phase:**The magic happens when CK goes high. The pre-charging transistors shut off, and C_{bat} is left floating with its V_{DD} charge trapped inside. Immediately, the bottom plate of C_{bat} is snapped to the analog input signal (V_{in}).Because the voltage across a capacitor can't jump instantly, the top plate is forced to "boost" up to $V_{in} + V_{DD}$. I then route this boosted voltage directly to the gate of M_{sw}. If you look at the math, the gate is at $V_{in} + V_{DD}$ and the source is at V_{in}, which means my V_{GS} is now exactly V_{DD}. It doesn't matter if my input is at 0.1V or 0.9V; the switch "sees" the same V_{GS} every single time.

4.2 Dynamic Comparator

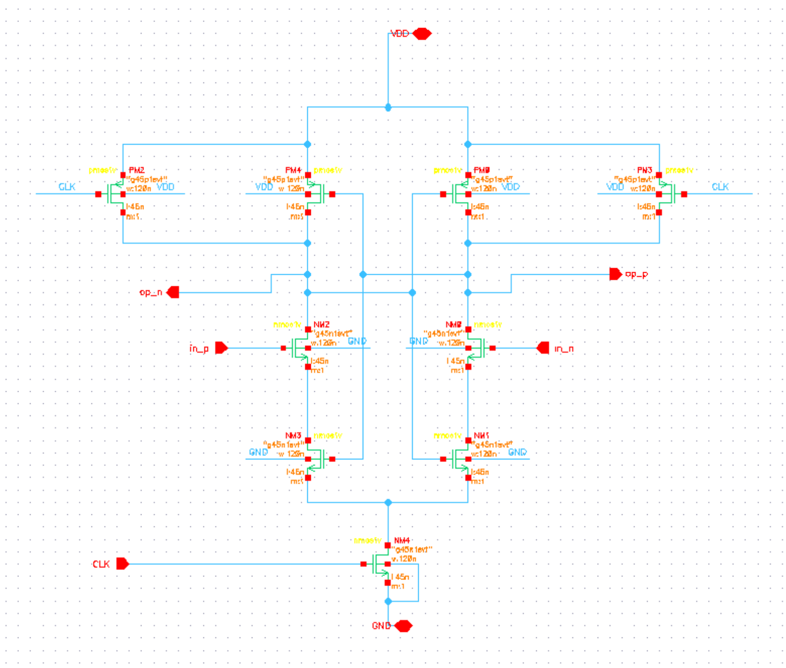

Figure 9: Single tail latch dynamic comparator

The comparator is the bridge between the analog DAC and the digital SAR logic. In most high-efficiency designs, we use a Single tail latch dynamic comparator. This setup is great because it isolates the sensitive input stage from the noisy output and uses a two-phase cycle (reset and evaluate) to ensure there's zero static power drain. When it's time to evaluate, the input transistors discharge internal nodes at slightly different rates depending on ΔV_{in}.Once that small difference is established, positive feedback kicks in to snap the signal to full logic levels. In an asynchronous flow, this behavior is the "heartbeat" of the system. Instead of waiting for a slow master clock, the comparator monitors its own outputs and fires a "Ready" pulse the moment it resolves, which instantly triggers the next step in the SAR logic.For biomedical uses, the biggest headache is flicker $(1/f)$ noise and DC offset. These are massive at the low frequencies where biosignals live. We handle this with chopper stabilization essentially swapping the inputs and outputs at a high frequency. This moves the signal away from the noise during processing. When we demodulate it back down, the noise gets pushed to high frequencies where a digital filter can easily kill it, keeping the signal-to-noise ratio high enough for clinical accuracy.

4.3 SAR Logic

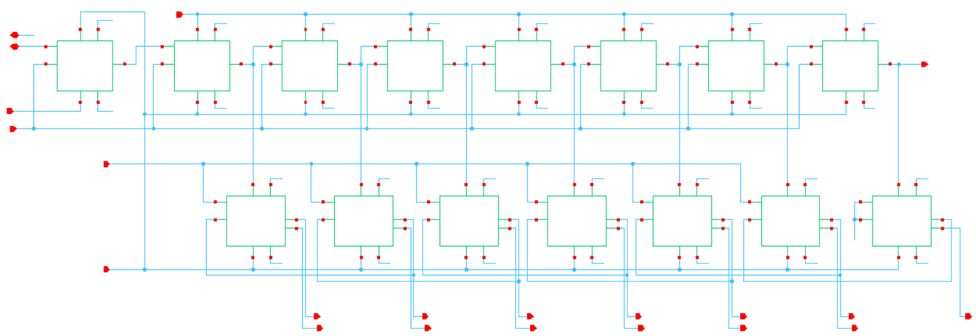

Figure 10: Basic block diagram of SAR Logic

I organized the overall SAR control logic into two distinct operational rows: a ring counter sequencer on the top and a data register on the bottom. The top row initializes a single logic high which I treat as a "token"—and shifts it one position to the right with every pulse from the comparator. This token acts as a moving spatial pointer that tells the system exactly which bit we are currently resolving. Specifically, the token enables one specific flip-flop in the bottom data row at a time. This enabled flip-flop then latches the comparator's output for that bit and immediately drives the corresponding switches in my capacitive DAC. This prepares the analog voltage for the next cycle of the binary search. By splitting the sequencer and the data register this way, I've created a clean, reliable path for the bit-by-bit approximation to occur in a self-timed, asynchronous loop.

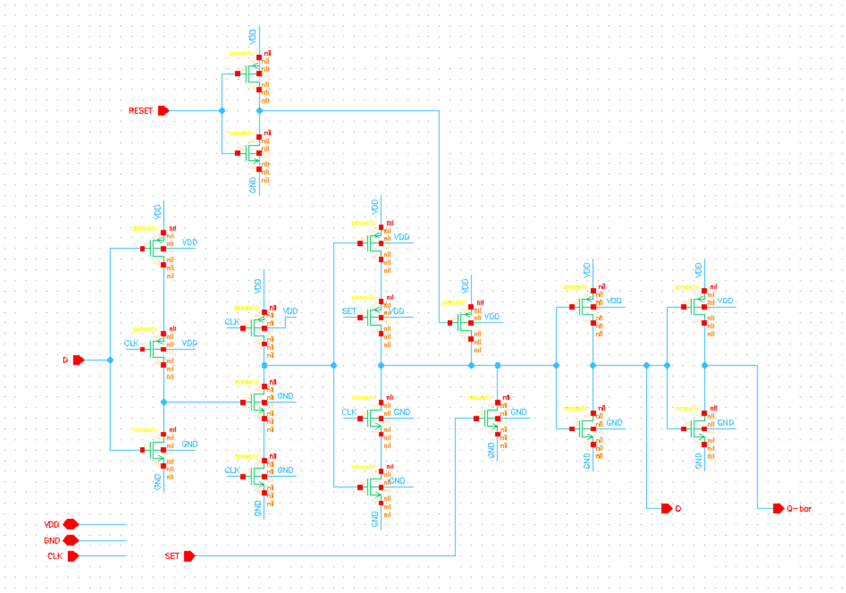

Figure 11: True single phase clock D flip flop

For the core of my digital system, I chose the True Single-Phase Clock (TSPC) D-flip-flop. This dynamic logic topology is a perfect fit for the tight timing requirements of my asynchronous flow. Unlike traditional static CMOS flip-flops that need power-hungry, two-phase clocks, the TSPC operates entirely on a single clock edge. In my framework, this "clock" is actually the self-timed "Ready" pulse from the comparator. The mechanism relies on dynamically storing charge on parasitic capacitances across three stages: a precharge input, an evaluation latch, and an output driver. When the comparator resets and the pulse is low, the internal nodes precharge to V_{DD}. The moment the comparator fires and the pulse goes high, the input stage evaluates the data, either discharging the internal node or holding the charge to push the logic state through. Because I've eliminated local clock inversion and reduced the transistor count, I've minimized the total switched capacitance. This directly impacts my power model, $P_{dyn} = \alpha C_L V_{DD}^2 f_{clk}$. By cutting the clock load (C_L), I can hit aggressive conversion rates without a massive power penalty.

4.4 capacitive DAC

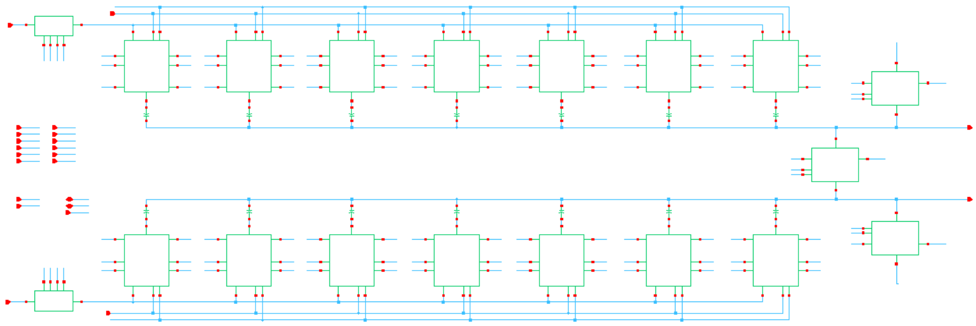

Figure 12: Basic block diagram of capacitive DAC

A capacitive DAC (CDAC) uses a binary-weighted capacitors array $128C$ to $1C$ capacitor array plus a small dummy capacitor- to convert digital bits into an analog voltage through charge redistribution. Each bit controls a switch that connects its capacitor either to the reference voltage V_{Ref} or the groung. When these switches toggles, charge moves among the capacitors, and the DAC output voltage is simply the final voltage at the array's common node. Because the array stores charge, it naturally acts as a sample-and-hold element, so no separate S/H circuit required.

Step-by-Step working principle

1. **Sampling Phase:** During smapling, all capacitor bottom plate connect to the input signal while the top node is grounded. This charges every capacitor to the input level. When sampling ends, the input is disconnected and the bottom plates are pulled to gound, causing the top node to shift to -Vin due to charge conservation.

2. **Conversion Phase:** The SAR Logic revolves bits from MSB to LSB. First, the largest capacitor is switched to Vref, shifting the common node upward by Vref/2. The comparator keeps this new voltage is above or below of Vref. If the input was larger than Vref, the comparator keeps the bits as '1'; otherwisw, It resets it to '0'. Each smaller capacitor is tested the same way, adding or removing binary-weighted fraction of Vref (Vref/4, Vref/8, etc.). With every step, the DAC effectively substracts scaled portion of Vref from the initially stored-Vin.

   After N steps, the charge redistributioin ideally bring the common node close to 0V. The final SAR register bits represent the digital value of the sampled input.

4.5 Output Register

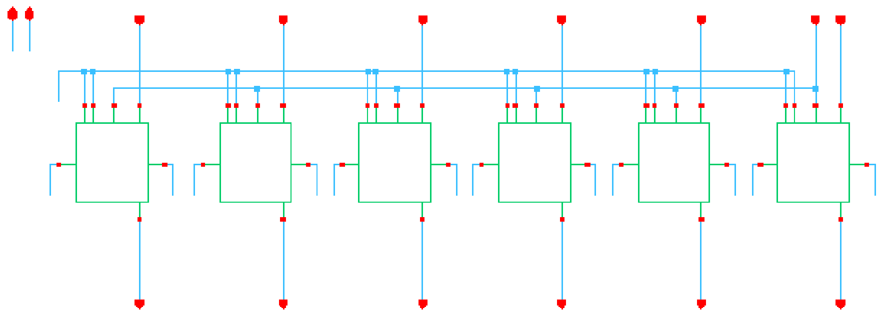

Figure 13: Basic block diagram of output Register

The output register serves as the final stage of the ADC, ensuring that only the completed conversion result is transferred to the digital domain. It captures the 8-bit DAC code exactly at the rising edge of the EOC signal, holding the final value while the next conversion begins. Once the EOC signal falls aligned with the rising edge of the next external clock the register releases the stored output, guaranteeing a clean and stable transition to the following digital circuitry.

The register is implemented using CMOS D-type flip-flops arranged in an 8-bit structure, as shown in the schematic. CMOS flip-flops are chosen because they provide static and reliable data retention without depending on dynamic charge storage. This ensures that the output remains valid for the entire duration of the hold phase, free from leakage or timing drift. Their simple clocking behavior also makes them well suited for the EOC-controlled latching mechanism, providing a robust and noise-free interface between the SAR logic and external digital blocks.

Chapter 5

Results and Discussion

5.1 Sample and Hold Circuit

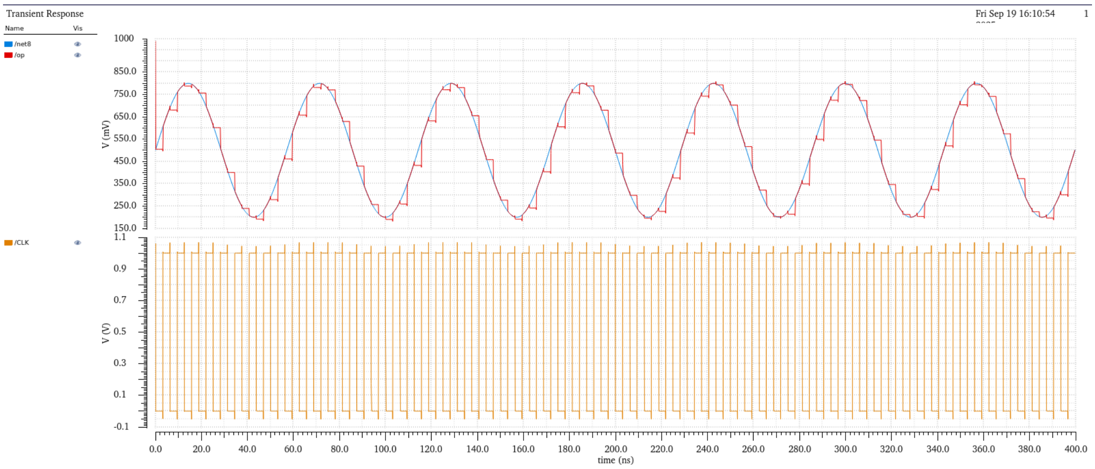

Figure 14: Sample & Hold Circuit waveform

I like to think of the Sample and Hold (S/H) circuit as a high-speed camera for electrical signals. Its main job is to take a "snapshot" of a fast-moving analog signal and hold it perfectly still so the rest of the system has enough time to figure out its value. Looking at the output in Fig. 14, you can see this in action: the blue line is the original signal moving up and down, while the red line is the "staircase" of snapshots we've taken. By looking at the orange clock signal at the bottom, we can see that the circuit is working incredibly fast, taking a new photo every 6.25ns. This means it's sampling at a frequency of 160 MHz, which is fast enough to capture all the important details of the input without missing a beat.To get those snapshots to look so clean, the hardware in Fig. 8 uses a clever trick called "bootstrapping". Inside the schematic, there is a small capacitor labeled C_0 that basically acts like a tiny, rechargeable battery. When the clock is low, the circuit "charges" this battery up to the supply voltage (V_{DD}). When the clock flips to high to take a sample, the circuit connects this "battery" directly to the gate of the main switch. This ensures the switch stays open at the exact same "width" every single time, regardless of whether the input signal is high or low. Without this trick, the switch would struggle to stay open for higher voltages, which would distort the signal and ruin the accuracy of our 8-bit digital result.For this to work perfectly, the circuit has to be tuned just right so that the "grab" happens almost instantly and the "hold" doesn't leak away. This is why the sizing of the transistors and the capacitor in the schematic is so critical; they have to be large enough to let the signal through quickly, but small enough to not interfere with the rest of the chip. By using a 1V supply and this bootstrapped architecture, I've ensured that the signal captured in the red staircase waveform is a near-perfect replica of the blue input. This gives the rest of the SAR ADC a rock-steady foundation to do its work, ensuring that every digital bit we resolve is based on a clean, stable reference point.

5.2 Dynamic Comparator

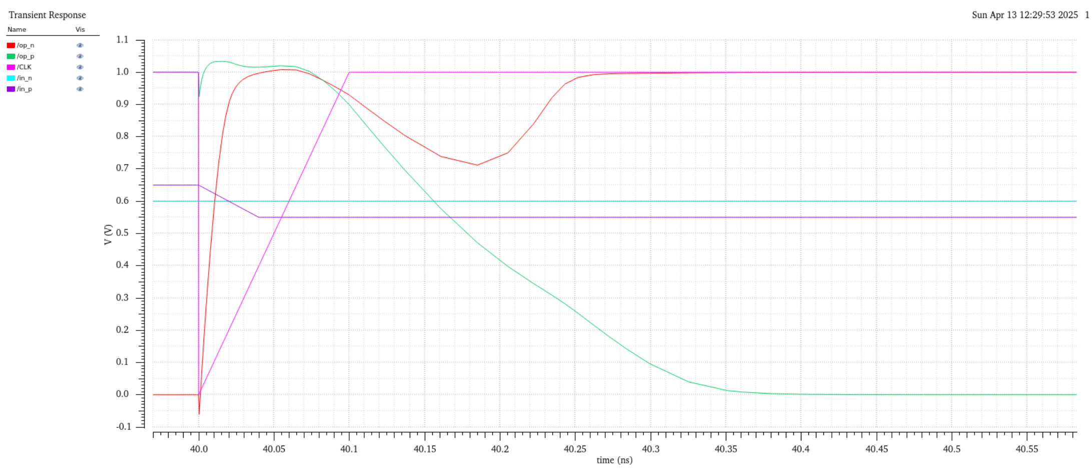

Figure 15: Comparator waveform

The comparator is a Single-Tail dynamic regenerative design. Zero static power draw, since it only dissipates energy during the brief evaluation phase. That alone makes it the obvious choice for a low-power 45nm build. It runs in two phases off CLK. When CLK is low, PM0 and PM3 precharge the internal nodes and pull both outputs (op_p and op_n) up to VDD. Just before the 40ns mark in the transient plot, both outputs are parked at 1.0V doing nothing. Then CLK goes high. The precharge transistors switch off, tail transistor NM4 opens a path to ground, and the comparison actually starts. I forced a tight input difference to stress-test it: in_n at 0.6V, in_p at 0.55V. The side driven by in_n sinks more current and its internal node drops faster. The cross-coupled inverters pick up that small head start and push positive feedback until the whole thing snaps. Between 40.1ns and 40.35ns you can watch the outputs fight neither has committed. Then the latch settles: op_p (green) hits 0V, op_n (red) climbs back to 1.0V. Roughly 250ps to resolve a 50mV difference. That's fast enough that the outputs are at full logic levels before the next clock edge, which means the SAR logic always sees a clean decision rather than a half-resolved one.

5.3 SAR logic

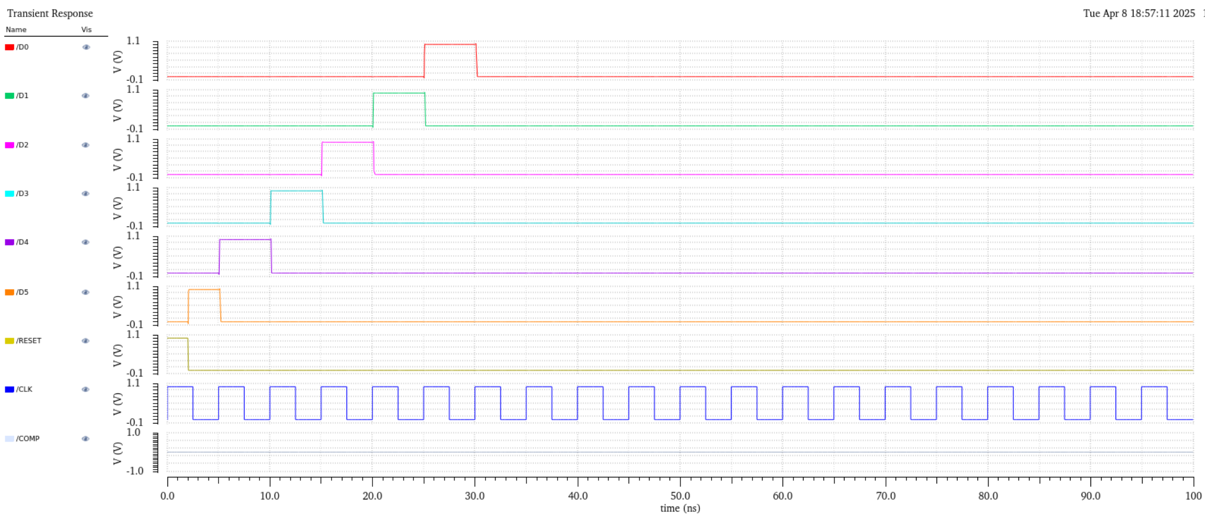

Figure 16: SAR logic waveform

The SAR control logic is what sequences the whole conversion. I used a two-row layout: a sequencer on top, a data register below. A Reset pulse clears the data registers and loads a single logic-high token into the first flip-flop of the sequencer. Every rising edge of clk_sar shifts that token one step right. Wherever the token sits, that's the one data flip-flop allowed to latch the comparator's output. One bit per cycle, MSB down to LSB. The transient response shows it working exactly as expected D7 pulses first, then the sequence steps down to D0, each bit resolved in a single 5ns cycle. Once all 8 bits are in, the logic fires an EOC pulse and the digital word is ready to read out. The part that needed the most care was timing. The logic has to settle and update the capacitive DAC fast enough that the comparator still has a useful window to evaluate before the next edge. If T_{logic} crept past the clock period, the sequence would lose sync and the conversion would be garbage.The TSPC flip-flops have enough headroom that this hasn't been a problem but it's the kind of thing that would bite you quietly if you weren't watching the delays.

5.4 Capacitive DAC

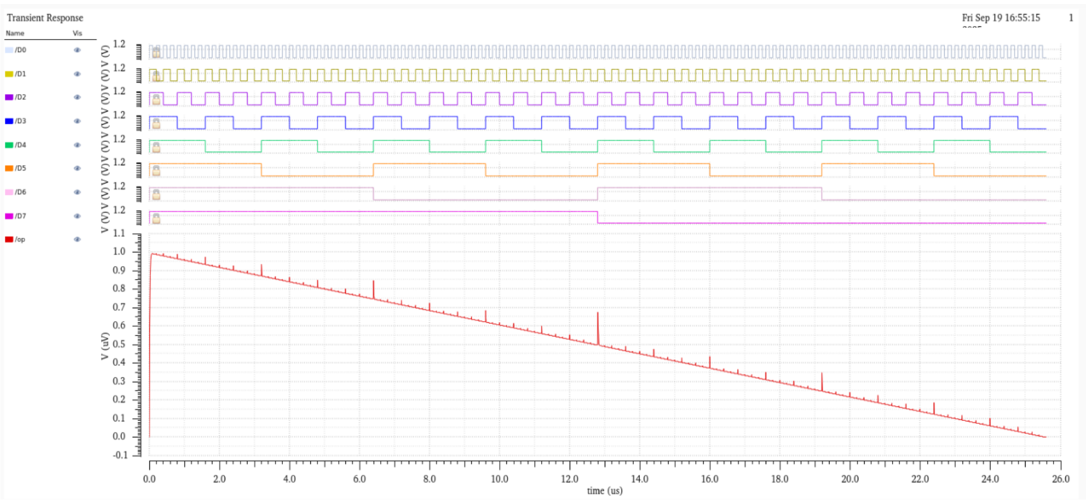

Figure 17: Capacitive DAC waveform

The working of the capacitive digital-to-analog converter (CDAC) in my design relies on a passive charge-redistribution architecture, which serves as the decision-making engine for the 8-bit SAR ADC. The process begins with the sampling phase, where the analog input signals, V_{INp} and V_{INn}, are captured through high-linearity bootstrapped switches. During this phase, the top plates of the binary-weighted capacitor array connected to the V_{XP} and V_{XN} nodes are tied to a common-mode voltage (V_{CM}), while the bottom plates track the input. My 8-bit array is structured with capacitors scaling from $128C$ for the most significant bit (MSB) down to a unit capacitor C, with a dummy unit capacitor included to ensure the total capacitance of the array is a perfect power of two for binary search accuracy. Once the sampling is complete and the input switches open, a specific charge representing the input signal is trapped on these nodes, governed by the principle of charge conservation.

The conversion phase then proceeds as a bit-by-bit binary search managed by the SAR control logic. The digital logic toggles the bottom plates of the capacitors between the positive and negative reference voltages, V_{Refp} and V_{Refn}, based on the comparator's output. Starting with the MSB ($D7$), switching the bottom plate to a reference voltage forces the charge to redistribute across the array, shifting the voltage at the V_{XP} and V_{XN} nodes in discrete binary-weighted steps. This iterative refinement is clearly visible in the transient response, where the analog output /op descends in smaller and smaller increments as the digital bits $D7$ through $D0$ are resolved. The final digital word is determined by which capacitors remain connected to V_{Refp}, effectively nulling the differential voltage seen by the comparator.

A major design challenge I identified during simulation was the impact of switching timing on the integrity of the analog output. In a 45nm process, digital control signals exhibit finite and sometimes mismatched rise and fall times (T_{rise} and T_{fall}). As shown in the zoomed-in transient plot, when multiple bits transition simultaneously specifically at the $12.8\mu s$ mark a significant voltage glitch occurs in the DAC output. This spike happens because bits like $D7$ might fall slightly faster than others rise, causing a momentary state where the CDAC represents an incorrect voltage level. To maintain 8-bit precision, I had to ensure that the system timing allowed enough margin for these glitches to settle completely before the comparator made its next decision, reinforcing the importance of coordinating the analog settling time with the asynchronous digital logic.

5.5 SAR ADC

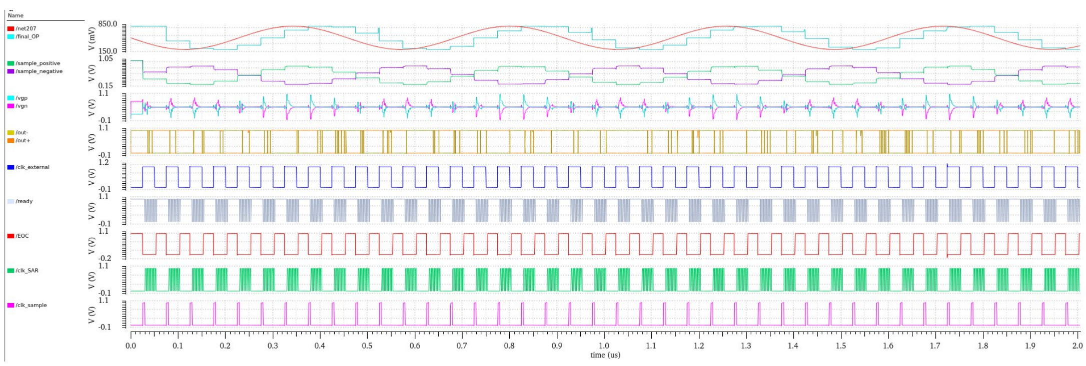

Figure 18: SAR ADC output

In my design, the timing of the entire SAR ADC is orchestrated by a specialized clock generator block that takes a master clk_external and derives the specific signals needed for each module: clk_sample, clk_SAR, and register_clk. The conversion process begins when clk_sample goes high, which triggers the bootstrapped switches in the DAC_TG block to trap the differential analog inputs onto the capacitor array. In the zoomed transient plot, I can see that once clk_sample falls, the register_clk also transitions to isolate the sampled voltages ($v_{inp\_sampled}$ and $v_{inn\_sampled}$), ensuring the analog signal is held steady before the intensive bit-by-bit comparison begins.

Once the sampling phase is locked, the system enters the iterative conversion phase driven by the clk_SAR signal. For each of the eight bits, clk_SAR triggers the dynamic comparator to evaluate the voltage difference between the differential DAC nodes, v_{gp} and v_{gn}. Every time the comparator successfully resolves a bit, it fires a ready pulse that serves as a self-timed handshake; this signal tells the TSPC-based SAR logic to latch the current bit and immediately switch the capacitive DAC bottom plates for the next approximation. You can actually see the internal latch nodes (latch+ and latch-) and the DAC voltages v_{gp} and v_{gn} gradually converging as the binary search narrows down the signal's value over the course of the clock cycles.

To achieve the high-fidelity reconstruction seen in the long-term transient, I had to manage the "settling time" as a primary working condition. The clk_SAR period must be long enough to allow the $128C$ to $1C$ capacitor array to redistribute its charge and settle to a stable voltage before the comparator triggers again. If the conversion is pushed beyond the 20 MHz limit without adequate settling margin, the LSB decisions would be based on incomplete voltages, leading to massive non-linearity. Once the eighth bit is decided, the register_clk pulses to latch the final digital word into the SAR_register, producing the final_OP staircase that accurately tracks the input sine wave. This seamless coordination between the self-timed comparator pulses and the rigid sampling clock is what allows the ADC to maintain 8-bit precision across the entire 1V input swing.

5.6 INL DNL PLOT

To evaluate the static linearity of my 8-bit SAR ADC, I generated the Differential Non-Linearity (DNL) and Integral Non-Linearity (INL) plots, which serve as the final verification of the converter's precision. My DNL plot shows a maximum deviation of $+0.650$ LSB and a minimum of -0.650 LSB. This result is particularly significant

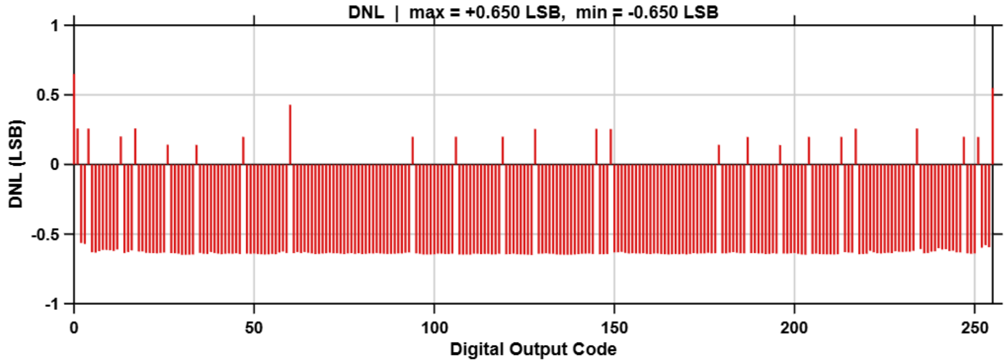

Figure 19: DNL Plot

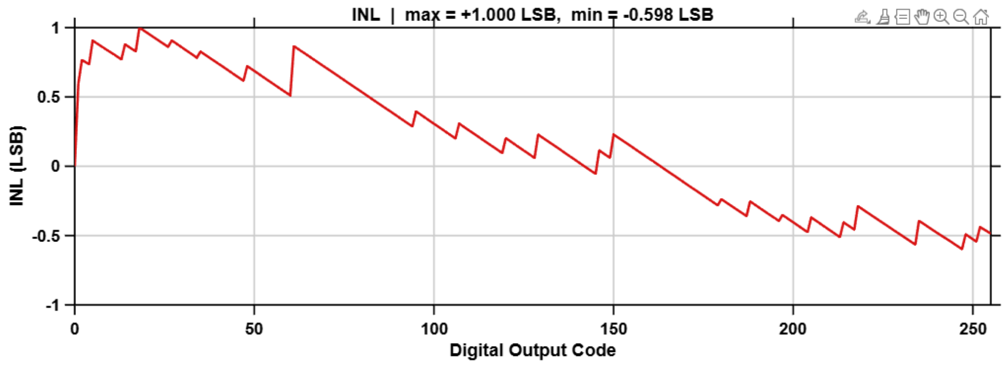

Figure 20: INL Plot

because it confirms that the ADC is monotonic; since the minimum DNL does not reach or exceed -1 LSB, there are no "missing codes" across the entire 256-code range. In a 45 nm process, keeping the DNL within this range is a direct result of careful capacitor sizing in the CDAC and ensuring that the switching transients from the TSPC logic settle completely before the comparator makes a decision.

The INL plot, which tracks the cumulative deviation from the ideal transfer characteristic, shows a peak positive error of $+1.000$ LSB and a minimum of -0.598 LSB. You can see a characteristic "sawtooth" pattern in the waveform, with sharp jumps occurring at major bit transitions—most notably at the mid-scale code of 128. These jumps are typical in binary-weighted capacitive DACs and are usually caused by capacitor mismatch or the parasitics in the routing. While a peak INL of 1 LSB is right at the boundary for a perfect 8-bit resolution, it remains within an acceptable threshold for low-power applications where the trade-off between silicon area and precision is critical.

5.7 Parameter

Parameter	**Value**
Technology	45nm CMOS
Power Supply	1 V
Resolution	8 Bits
Common-Mode Voltage	0.5 V
SINAD	43.4786 dB
ENOB	6.932 bits
Total Power	19.88 μW

Table 2: Performance Summary of the designed 8-bit SAR ADC.

Chapter 6

Conclusion

The successful design and simulation of a 8-bit SAR ADC architecture using 45nm CMOS technology highlight its feasibility and efficiency for low-power, medium-resolution applications. Each functional block—Sample-and-Hold, Comparator, SAR Logic, Capacitive DAC, and Output Register—was carefully optimized and tested to ensure reliable conversion with minimal energy consumption. The power-efficient comparator and streamlined SAR control logic ensure accurate and synchronized bit decisions, while the Capacitive DAC provides stable and rapid voltage steps for precise analog-to-digital conversion.

Simulation results confirm the expected behavior of all blocks and demonstrate smooth interaction in the integrated ADC system. Despite minor waveform imperfections, the overall performance adheres to design expectations, establishing a robust platform for further extension to higher resolutions or real-world implementation. This work lays a strong groundwork for energy-efficient ADC solutions in biomedical, portable, and embedded applications.

References

[1] ham, X.T., Nguyen, T.V.H., Le, V.T., Tran, T.T.H., Phung, T.N. and Hoang, M.K., 2023, December. A Low Noise 10-Bit 59.41 dB SNR SAR ADC Using Chopper Comparator for Biomedical Applications. In International Conference on Advances in Information and Communication Technology (pp. 18-25). Cham: Springer Nature Switzerland.

[2] ang, J., 2023, November. A 10-bit 160 MS/s Asynchronous SAR ADC design. In Journal of Physics: Conference Series (Vol. 2645, No. 1, p. 012001). IOP Publishing.

[3] amalinejad, P., Mirabbasi, S. and Leung, V.C., 2011, May. An ultra-low-power SAR ADC with an area-efficient DAC architecture. In 2011 IEEE International Symposium of Circuits and Systems (ISCAS) (pp. 13-16). IEEE.

[4] hiue, M.T., Lo, Y.F. and Jung, C.Y., 2024. Design of an internal asynchronous 11-bit SAR ADC for biomedical wearable application. Electronics, 13(17), p.3549.

[5] itnaware, S.R., Tijare, A.D. and Matey, M.M., 2015, April. Design of binary architecture for successive approximation analog-to-digital converter. In 2015 International Conference on Communications and Signal Processing (ICCSP) (pp. 0512-0516). IEEE..